

PLEVEN, Bulgaria

WMO: 155280

Lat: 43.400N

Lon: 24.600E

Elev: 160

StdP: 99.42

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.6	-14.9	1.0	-11.9	-11.8	1.4	-8.7	14.5	5.8	12.3	5.7	1.0	70	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	34.7	22.0	32.8	21.8	31.2	21.2	23.8	31.6	22.8	30.4	21.9	29.3	2.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.2	28.4	20.2	15.2	27.1	19.3	14.4	25.9	72.0	31.8	68.2	30.4	64.9	29.4	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	9.9	7.7	-16.4	37.6	3.4	2.2	-18.8	39.2	-20.8	40.5	-22.8	41.7	-25.2	43.3		
(5)			-16.7	25.7	3.4	1.8	-19.1	27.0	-21.1	28.1	-23.1	29.1	-25.5	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	0.2	2.2	7.3	12.8	18.1	21.8	23.9	23.7	18.9	12.8	6.7	0.8
	DBStd	9.51	5.27	5.92	4.85	3.64	3.43	3.15	2.83	2.96	3.51	4.40	4.50	5.01
	HDD10.0	1087	306	221	112	16	1	0	0	0	23	120	289	
	HDD18.3	2688	562	451	342	169	47	7	1	2	34	177	351	545
	CDD10.0	1992	2	4	28	99	251	353	431	426	267	110	20	2
	CDD18.3	553	0	0	0	2	39	110	173	169	52	6	1	0
	CDH23.3	5406	0	0	3	41	364	1048	1749	1671	466	62	2	0
	CDH26.7	2181	0	0	0	8	89	396	782	744	150	12	0	0
Wind	WSAvg	2.3	2.2	2.7	3.2	2.7	2.3	2.3	2.2	2.0	2.0	1.9	2.2	2.0
Precipitation	PrecAvg	584	42	37	38	51	66	78	60	47	41	33	49	42
	PrecMax	758	115	73	123	114	136	174	140	149	154	150	107	93
	PrecMin	405	0	3	2	1	6	16	5	2	2	1	1	3
	PrecStd	106	29	19	27	28	31	45	34	39	39	30	28	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	19.0	23.2	27.6	31.6	35.0	37.3	37.5	33.7	28.2	22.1	16.5
		MCWB	9.2	11.8	12.5	17.0	20.5	22.5	21.8	21.7	20.9	19.1	14.8	11.1
	2%	DB	12.1	15.9	20.4	23.7	29.0	32.6	34.8	34.6	30.4	24.9	18.8	12.5
		MCWB	7.4	9.9	11.9	15.5	19.2	22.1	21.9	22.1	19.6	17.3	13.5	9.0
	5%	DB	9.7	13.4	17.9	21.5	27.2	30.8	32.7	32.7	28.1	22.3	15.5	9.3
		MCWB	6.5	8.5	10.5	14.0	18.4	21.7	21.9	21.6	18.6	16.0	11.5	6.8
	10%	DB	7.3	10.4	15.4	19.7	25.3	29.0	31.0	30.9	25.9	19.9	13.1	7.0
		MCWB	4.9	7.0	9.5	12.8	17.7	20.8	21.3	20.8	17.7	14.9	10.3	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	12.2	14.2	18.1	21.6	24.6	25.2	25.9	22.4	20.2	15.6	11.3
		MCDB	14.2	17.5	20.7	26.5	29.6	31.6	32.7	32.6	30.6	27.1	20.2	14.9
	2%	WB	7.9	10.3	12.4	16.1	20.3	23.2	23.6	23.8	20.6	17.9	13.7	9.1
		MCDB	11.3	15.4	18.8	22.6	27.4	30.6	31.6	31.7	28.5	23.6	18.0	12.2
	5%	WB	6.6	8.7	11.1	14.7	19.2	22.2	22.6	22.6	19.4	16.4	12.2	7.2
		MCDB	9.5	12.8	16.8	20.3	25.6	29.4	30.6	29.6	26.5	21.4	14.8	9.2
	10%	WB	5.1	7.1	9.7	13.5	18.2	21.2	21.8	21.4	18.4	15.1	10.7	5.5
		MCDB	7.3	10.2	15.1	18.6	24.2	27.9	29.4	28.8	24.5	19.3	12.9	6.9
Mean Daily Temperature Range	5% DB	MDBR	6.2	8.0	9.4	10.3	10.9	11.2	11.6	11.8	10.8	9.6	7.1	5.9
		MCDBR	10.0	13.5	13.9	14.0	14.2	14.3	14.8	15.1	14.8	13.8	11.8	9.9
		MCWBR	6.7	8.6	7.7	7.5	6.7	6.3	6.0	5.7	6.3	7.5	7.3	6.7
	5% WB	MCDBR	9.6	12.9	13.4	13.0	12.9	13.4	13.3	13.4	13.6	13.0	10.7	9.1
		MCWBR	6.8	8.5	8.2	7.6	7.0	6.8	6.5	6.6	6.9	7.6	7.0	6.4
Clear-Sky Solar Irradiance	taub	0.337	0.367	0.408	0.464	0.443	0.447	0.456	0.467	0.436	0.395	0.362	0.341	
	taud	2.314	2.241	2.179	2.034	2.165	2.193	2.173	2.139	2.197	2.296	2.353	2.333	
	Ebn at Noon	775	811	823	803	833	830	816	791	785	770	741	732	
	Edh at Noon	85	108	130	162	146	143	144	145	127	101	82	77	
All-Sky Solar Radiation	RadAvg	1.48	2.31	3.46	4.73	5.81	6.41	6.50	5.82	4.24	2.77	1.60	1.23	
	RadStd	0.22	0.30	0.33	0.49	0.48	0.41	0.40	0.39	0.33	0.33	0.22	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.73	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		+0.47	N/A	N/A	N/A	+0.42	N/A	N/A	-142	+70	N/A

Nomenclature: See separate page